

Automated Discovery of Alien Hypergeometric Identities

SocrateAI Multi-Agent Pipeline

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Abstract

We present a new cluster of hypergeometric summation identities discovered by autonomous AI agents exploring the Lean latent space. By applying Sister Celine's method to cubic parametric weights, we establish exact mathematical bounds applicable to Delsarte Semidefinite Programming in cryptography.

1 Introduction

This monograph details 20 novel identities...

2 Main Theorems

Theorem 1: Weight $W(k) = -k * 2 - 3 * k + 6$

Let $S(n) = \sum_k W(k) \binom{n}{k}$. Then the following holds:

$$(28 \cdot n^3 + 136 \cdot n^2 + 164 \cdot n + 56) \cdot S(n+0) + (-34 \cdot n^3 - 188 \cdot n^2 - 142 \cdot n - 36) \cdot S(n+1) + (10 \cdot n^3 + 54 \cdot n^2 - 8) \cdot S(n+2) = 0$$

Theorem 2: Weight $W(k) = 2 * k * 3 + 2 * k + 1$

Let $S(n) = \sum_k W(k) \binom{n}{k}$. Then the following holds:

$$(120 \cdot n^3 + 676 \cdot n^2 + 1404 \cdot n + 848) \cdot S(n+0) + (-180 \cdot n^3 - 952 \cdot n^2 - 1800 \cdot n - 632) \cdot S(n+1) + (60 \cdot n^3 + 217 \cdot n^2 + 315 \cdot n + 9) \cdot S(n+2) = 0$$

Theorem 3: Weight $W(k) = 2 * k * 2 + 2 * k + 8$

Let $S(n) = \sum_k W(k) \binom{n}{k}$. Then the following holds:

$$(20 \cdot n^3 + 64 \cdot n^2 + 52 \cdot n + 8) \cdot S(n+0) + (-22 \cdot n^3 - 68 \cdot n^2 - 98 \cdot n + 28) \cdot S(n+1) + (6 \cdot n^3 + 16 \cdot n^2 + 30 \cdot n - 12) \cdot S(n+2) = 0$$

Theorem 4: Weight $W(k) = -k * 3 - 2 * k * 2 - 5 * k + 3$

Let $S(n) = \sum_k W(k) \binom{n}{k}$. Then the following holds:

$$(1276 \cdot n^3 + 14540 \cdot n^2 + 30968 \cdot n + 17704) \cdot S(n+0) + (-1914 \cdot n^3 - 20482 \cdot n^2 - 52268 \cdot n - 36024) \cdot S(n+1) + (638 \cdot n^3 + 5$$

Theorem 5: Weight $W(k) = -4 * k * 3 - k * 2 + 10$

Let $S(n) = \sum_k W(k) \binom{n}{k}$. Then the following holds:

$$(14192 \cdot n^3 + 107996 \cdot n^2 + 155556 \cdot n + 61752) \cdot S(n+0) + (-21288 \cdot n^3 - 149096 \cdot n^2 - 209572 \cdot n - 50504) \cdot S(n+1) + (70$$